If we call the input number x and the output number y, we can then write the 'rule' in terms of x and y.

$x \rightarrow$ [multiply by 3] $\rightarrow$ [add 2] $\rightarrow y$

The rule for this function machine is 'multiply by 3, then add 2'.

We can write this as $x \times 3 + 2 = y$ or $y = 3x + 2$.

Here are the rules for these function machines:

$x \rightarrow$ [multiply by 2] $\rightarrow$ [add 4] $\rightarrow y$ Rule: $y = 2x + 4$

$x \rightarrow$ [multiply by 8] $\rightarrow$ [subtract 7] $\rightarrow y$ Rule: $y = 8x - 7$

The rule $y = 2x + 4$ may also be written as $\boldsymbol{x \rightarrow 2x + 4}$.

Exercise 21.1

1. Find the output for each of these:

(i) 8 $\rightarrow$ [subtract 6] $\rightarrow$ ...

(ii) 4 $\rightarrow$ [multiply by 4] $\rightarrow$ ...

(iii) 5 $\rightarrow$ [multiply by 2] $\rightarrow$ [add 1] $\rightarrow$ ...

(iv) 3 $\rightarrow$ [multiply by 3] $\rightarrow$ [subtract 4] $\rightarrow$ ...

(v) 6 $\rightarrow$ [divide by 2] $\rightarrow$ [add 3] $\rightarrow$ ...

(vi) 10 $\rightarrow$ [subtract 5] $\rightarrow$ [multiply by 3] $\rightarrow$ ...

2. Describe in **words** the rule for each function machine in Question **1.** above.

3. Find the outputs for these:

(i) 3, 7, 8 $\rightarrow$ [multiply by 2] $\rightarrow$ [subtract 4] $\rightarrow$ ..., ..., ...

(ii) 2, 1, 0 $\rightarrow$ [add 4] $\rightarrow$ [multiply by 2] $\rightarrow$ ..., ..., ...

4. $x \rightarrow$ [add 2] $\rightarrow$ [multiply by 3] $\rightarrow y$ is a function machine.

Use a copy of this table and fill in the values for y.

x	1	2	3	4	5
y					

Write in words what this function machine does.

5. Write the rules for these function machines as $y = \ldots$

(i) $x \rightarrow$ [multiply by 2] $\rightarrow$ [add 6] $\rightarrow y$

(ii) $x \rightarrow$ [multiply by 8] $\rightarrow$ [subtract 9] $\rightarrow y$

(iii) $x \rightarrow$ [divide by 4] $\rightarrow$ [subtract 3] $\rightarrow y$

(iv) $x \rightarrow$ [add 3] $\rightarrow$ [multiply by 4] $\rightarrow y$

6. Write the rules for these function machines in the form $x \rightarrow \ldots$

(i) $x \rightarrow$ [multiply by 3] $\rightarrow$ [add 2] $\rightarrow$

(ii) $x \rightarrow$ [multiply by 5] $\rightarrow$ [subtract 2] $\rightarrow$

(iii) $x \rightarrow$ [divide by 3] $\rightarrow$ [subtract 6] $\rightarrow$

(iv) $x \rightarrow$ [double the number] $\rightarrow$ [add 7] $\rightarrow$

7. Find the missing operation in each of these:

(i) 3, 5, 8 $\rightarrow$ [?] $\rightarrow$ 15, 25, 40

(ii) 9, 1, 15 $\rightarrow$ [?] $\rightarrow$ 13, 5, 19

(iii) 24, 16, 100 $\rightarrow$ [?] $\rightarrow$ 6, 4, 25

8. Find the missing operations in these function machines:

(i) 3, 2, 1 $\rightarrow$ [multiply by 2] $\rightarrow$ [?] $\rightarrow$ 5, 3, 1

(ii) 0, 1, 5 $\rightarrow$ [multiply by 3] $\rightarrow$ [?] $\rightarrow$ 1, 4, 16

(iii) 4, 1, 6 $\rightarrow$ [multiply by 3] $\rightarrow$ [?] $\rightarrow$ 10, 1, 16

(iv) 1, 2, 3 $\rightarrow$ [?] $\rightarrow$ [multiply by 4] $\rightarrow$ 12, 16, 20

9. What numbers went into each of these function machines?

(i) ?, ?, ? $\rightarrow$ [multiply by 3] $\rightarrow$ 9, 15, 21

(ii) ?, ?, ? $\rightarrow$ [subtract 4] $\rightarrow$ 6, 12, 9

(iii) ?, ?, ? $\rightarrow$ [multiply by 2] $\rightarrow$ [add 1] $\rightarrow$ 11, 17, 41

(iv) ?, ?, ? $\rightarrow$ [multiply by 3] $\rightarrow$ [subtract 1] $\rightarrow$ 11, 23, 29